

Wetlands, Flooding, and the Clean Water Act[†]

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In 2020 the Environmental Protection Agency narrowed the definition of “waters of the United States,” significantly limiting wetland protection under the Clean Water Act. Current policy debates center on the uncertainty around wetland benefits. We estimate the value of wetlands for flood mitigation across the United States using detailed flood claims and land use data. We find the average hectare of wetland lost between 2001 and 2016 cost society \$1,840 annually, and over \$8,000 in developed areas. We document significant spatial heterogeneity in wetland benefits, with implications for flood insurance policy and the 50 percent of “isolated” wetlands at risk of losing federal protection. (JEL K32, Q24, Q25, Q53, Q58)

The Clean Water Act (CWA) is among the most important, wide-reaching, and costly pieces of environmental legislation in the United States. A primary component of the CWA is the regulation of wetlands under Section 404, which requires a permit in order to dredge or fill aquatic resources for real estate, infrastructure, mining, or industrial activities. In June 2020, the Navigable Waters Protection Rule (NWPR) narrowed the extent of “waters of the United States” (WOTUS) that fall under CWA jurisdiction. Under this rule, an estimated 50 percent of wetland area may no longer be protected under Section 404 (Sullivan, Rains, and Rodewald 2019). A subject of long-standing legal debate, federal jurisdiction over wetlands is the focus of the upcoming Supreme Court case, *Sackett v. Environmental Protection Agency*. The stakes are high: Section 404 is perhaps the most important federal regulation governing land use with major implications for zoning regulations, urban development, and consequently aggregate economic growth (Hsieh and Moretti 2019).

Underlying this debate is the highly controversial nature of wetland regulation, which pits the protection of productive ecosystems against the rights of private landowners. Wetlands provide a multitude of public goods in the form of ecosystem services, including flood mitigation, water quality improvement, and wildlife habitat.

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Many of these services provide off-site benefits, which the resource owner is unable to capture (Turner et al. 2000). There is limited financial incentive to preserve wetlands since the private benefits derived by the landowner do not reflect the full benefits of wetlands to society (Heimlich, Claassen, and Wiebe 1998).

Wetland regulation is intended to correct this market failure. While the costs of regulation are well documented, the benefits are more difficult to quantify, making benefit-cost analysis challenging (Turner, Georgiou, and Fisher 2012). A common concern is that policy-makers might undervalue resources that are not quantified, impeding proper management (Costanza et al. 2014). This issue is central to the debate over the recent NWPR, which excludes the value of ecosystem services provided by wetlands from its benefit-cost analysis, citing a lack of credible empirical evidence on the economic value of wetlands for flood mitigation and other ecosystem services.

This study estimates the value of wetlands for flood mitigation in the United States. Although flood mitigation is only one of many ecosystem services provided by wetlands, rich data on flood damages allows for a uniquely tractable empirical evaluation of the contribution of wetlands to human well-being. We conduct a large-scale analysis across the entire continental US linking zip code-level data on the Federal Emergency Management Agency's (FEMA's) National Flood Insurance Program (NFIP) claims with high-resolution data on wetland area from 2001 to 2016.

To our knowledge, we are the first to estimate the causal effect of wetlands on flood damages. There is limited empirical evidence on the economic value of these ecosystems for flood mitigation, particularly for freshwater wetlands, which represent 95 percent of total wetland area in the United States (Brander, Florax, and Vermaat 2006; Barbier 2013; Dahl 2011). Analyzing 34 hurricanes, Costanza et al. (2008) find coastal wetland to be associated with an average \$8,000 per hectare reduction in annual storm damage in the United States, a general finding supported by recent papers (Narayan et al. 2017; Sun and Carson 2020). However, these studies do not explicitly assert or test causal mechanisms. Importantly, prior work focuses primarily on coastal wetlands and damages from hurricanes, and may not be representative of the vast majority of US wetlands, which are located inland, or more typical flood events, which, unlike hurricanes, are not accompanied by storm force winds.

The empirical challenge is that wetland spatial extent is associated with other factors that drive flood damage dynamics. For example, locations with wetter climates are more likely to experience flood events and also have more wetlands. Additionally, locations experiencing rapid population growth are likely to see both reductions in wetland area due to urban expansion into natural areas and higher flood claims due to an increase in the housing stock or changes in flood insurance uptake rates. Such omitted variables may confound estimates of the economic value of wetlands for flood mitigation. To address this challenge, we employ three different identification strategies—long differences, panel analysis, and upstream-downstream difference-in-differences—which rely on different assumptions and identifying variation to establish a causal effect. These methods yield quantitatively similar results, providing plausible bounds for the causal effect of wetlands on flood damages.

We estimate that each hectare of wetland loss between 2001 and 2016 increases NFIP claims by \$1,840 to \$1,900 per year when accounting for spatial spillovers. However, this value masks significant spatial heterogeneity in wetland benefits. For example, we estimate that one hectare of wetland loss in developed areas (those with over 10 percent built-up area) costs society \$8,290 in flood mitigation value. We show that the benefits of wetlands come from their protective services in relation to their “sponge-like” capacity to absorb water, rather than real estate development occurring in the flood-prone areas that lost wetlands.

To put these values into context, we use high-resolution land price data (Nolte 2020) to estimate the value of private lands that are classified as wetlands. We find that the mean market value across all wetlands in the continental United States is \$12,700 per hectare, while wetlands that were lost between 2001 and 2016 have an estimated value of \$31,600 per hectare. Using this range, the societal benefits from reduced flooding outweigh the cost of conserving wetlands (based on land price) within 6 to 22 years, on average. One interpretation of our results is that lifting federal protections for wetlands represents a transfer from taxpayers, who fund the NFIP, to private landowners, who profit from converting wetlands to other uses.

We note that our estimates are a lower bound on the overall value of wetlands that exclude their benefits in relation to any flood mitigation occurring outside of the NFIP program (e.g., commercial, governmental, or otherwise privately insured properties, as well as farm assets and crop production). Additionally our estimates do not take into account the value of wetlands for recreation, habitat, water quality, and the fishing industry.

Our results have several important policy implications: first, we find no effect of *increases* in wetland area on flood insurance claims, calling into question the extent to which wetland restoration can offset the adverse impacts of wetland loss. Second, we find substantial spatial spillovers in the benefits of wetlands, supporting the often-cited theory that regulation may be required to achieve the optimal provision of wetlands due to the presence of positive downstream externalities. Third, we evaluate wetland location relative to the surface water network and find that the most valuable wetlands for flood mitigation are those located 500 to 750 meters from the nearest stream or river. This finding is at odds with the NWPR’s interpretation of WOTUS that eliminates federal protections for the estimated 50 percent of “isolated” wetlands that lack a surface water connection.

We also document significant heterogeneity in the impact of wetlands on flood damages. Geographically, we find the greatest benefits in the eastern half of the United States. We observe heterogeneity by ecoregion and type of land use conversion, which aligns with the fact that wetlands and their functions vary greatly by type, location, and position—suggesting that a more decentralized implementation of the CWA may be appropriate for wetland regulation. Finally, we find the flood mitigation benefits of wetlands to be greatest during anomalously high precipitation events, which are projected to become more frequent with climate change.

The remainder of the paper proceeds as follows: Section I provides background information on the relationship between wetlands and flooding, wetland regulation under the CWA, and the National Flood Insurance Program. Sections II to IV outline our data and identification strategies. Section V presents our empirical estimates of

the effect of wetlands on flood damages, along with several extensions. Section VI is a discussion of the implications of our findings.

I. Background

A. Wetlands and Flooding

Examples of recent extreme flooding in the US Midwest (2019); Houston (2017); Puerto Rico (2017); and Baton Rouge (2016) have underscored the human and economic cost of floods. The most common disaster globally, flooding is responsible for one-half of all natural disaster deaths. Floods were responsible for 6.8 million deaths in the twentieth century (Doocy et al. 2013) and impose a significant public health burden (Alderman, Turner, and Tong 2012). In 2019 floods caused \$82 billion in global economic losses, and cumulative losses since 2000 are estimated to be \$1 trillion (Aon 2019). In the United States, the Congressional Budget Office estimates that flooding causes \$20 billion in annualized economic losses.

Flood risk is expected to increase in the coming years with rising sea levels and more frequent and extreme precipitation events (Knutson et al. 2010). Consequently, there is increased demand for flood mitigation strategies that are physically sound, cost-effective, and politically feasible (Woodruff, Irish, and Camargo 2013). There is a lack of agreement on the best way to mitigate flood damages. While one effective approach is to relocate people and capital away from flood-prone areas, this option is often politically infeasible. The most common approach is environmental engineering (i.e., levees) but such practices generally displace the destructive force of floodwaters elsewhere.

An alternative strategy is nature-based solutions that utilize ecosystem functions of wetlands to reduce flood risk (Spalding et al. 2014). Wetlands are transitional lands between terrestrial and aquatic ecosystems that are water saturated enough to support hydrophytic vegetation and hydric soils. Wetlands mitigate flooding by acting as “natural sponges” that absorb and hold floodwaters until they are able to infiltrate the ground or slowly release into nearby streams. Wetland vegetation, such as trees and root mats, help reduce the speed of floodwaters moving across floodplains. In combination, these two properties of wetlands have been shown to influence the peak flows, volume, timing, and duration of floods (Acreman and Holden 2013; Thomas and Nisbet 2007). However, estimating the flood mitigation services of wetlands is far from straightforward given the complexity in hydrological connections between wetlands, streams, and rivers which varies by each location’s physical characteristics including its climate, soils, geology, topography, and spatial distribution of aquatic features (EPA 2015; Leibowitz et al. 2018).

B. Wetland Regulation under the CWA

The CWA was passed in 1972 with a stated purpose “to restore and maintain the chemical, physical, and biological integrity of the Nation’s waters.” A primary component of the CWA is Section 404, which prohibits any activity involving the discharge of dredged or fill material into WOTUS unless a permit is issued by the US Army Corps of Engineers (EPA 2020c) in cooperation with the Environmental

Protection Agency (EPA). Such regulated waters include wetlands, a land form encompassing over 5 percent of the continental United States (Dahl 2011). The law thus affects many segments of the economy: real estate development, infrastructure (i.e., highways, ports, airports), water resources (i.e., dams, levees), industrial development, agriculture, and mining.

Over the last several decades, the national policy goal has been one of “no net loss” of wetland area, prompting the development of compensatory mitigation requirements under CWA Section 404 (EPA 2020a). This program requires that when adverse impacts of wetlands are unavoidable, developers must offset the reduction in wetland functions by restoring, creating, or enhancing other wetlands in the watershed.

CWA Section 404 imposes financial and administrative burdens on many segments of the economy. The costs of the program are well documented: the EPA estimates that Section 404 costs developers between \$20 million and \$53 million per year in permit applications and compensatory mitigation costs (EPA 2013). In addition, the program carries administrative costs for the federal government estimated at \$7 to \$11 million annually, as well as opportunity costs associated with delays in project implementation (EPA 2013; Sunding and Zilberman 2002). In contrast, the benefits of wetland regulation, in the form of ecosystem services, are more difficult to quantify because they are not captured in the market and many of the benefits accrue to off-site users.

Legal debates over the CWA focus on the definition of WOTUS, which determines which areas receive federal protection under the legislation. WOTUS has been subject to several Supreme Court decisions over the decades.¹ Following an executive order from President Trump in 2017, the EPA and the Army Corps implemented the NWPR in June 2020, redefining WOTUS to exclude wetlands considered “isolated” from navigable waters (EPA and Army Corps 2020). Specifically, most wetlands lacking surface water connections to intermittent or perennial streams are no longer protected. Although it is unclear how this rule would be applied in practice, a 2017 analysis by EPA and the Army Corps estimates that over 50 percent of the nation’s wetlands would no longer be under federal jurisdiction (Sullivan, Rains, and Rodewald 2019).

The Supreme Court’s announcement in January 2022 that it intends to hear *Sackett v. Environmental Protection Agency* follows several lawsuits filed in federal court, challenging the new WOTUS rule. Plaintiffs include environmental groups like National Wildlife Federation and Natural Resources Defense Council,² as

¹ *United States v. Riverside Bayview Homes* (1985) upheld that wetlands adjacent to navigable waters were “inseparably bound” and to be included in the definition of WOTUS; *Solid Waste Agency of Northern Cook County v. US Army Corps of Engineers* (2001) found that isolated nonnavigable intrastate ponds utilized by migratory birds did not constitute a sufficient basis; *Rapanos v. United States* (2006) concluded that WOTUS encompassed some waters not navigable in the traditional sense. Justice Kennedy stated that to be covered by the CWA “a water or wetland must possess a ‘significant nexus’ to waters that are or were navigable in fact or that could reasonably be so made.” Significant nexus is achieved if the wetlands “either alone or in combination with similarly situated [wet] lands in the region, significantly affect the chemical, physical, and biological integrity of other covered waters more readily understood as ‘navigable.’”

² Available: https://www.southernenvironment.org/uploads/words_docs/2019.10.23_-_Final_Repeal_Rule_Complaint.pdf.

well as 17 states, Washington DC, and New York City.³ The legal debate centers on the extent to which WOTUS extends beyond an original narrow definition of navigable waters that encompassed tributaries (perennial, intermittent, and ephemeral) and other connected waters. Of particular concern is WOTUS's treatment of nonadjacent, or "isolated," wetlands.⁴

The previous definition of WOTUS, established in 2015, relied on Justice Kennedy's Rapanos criteria of "significant nexus" to determine what constitutes jurisdictional waters. In applying this guidance, the EPA and the Army Corps estimated a value of wetlands ranging from \$129,000 to \$292,000 per acre based on their combined fishing, hunting, fur trapping, recreation, water filtration, flood control, aesthetic, and habitat value (EPA 2013). Acknowledging potential bias due to double counting, nonrepresentative wetlands, and lack of empirical strategies to assert causal inference, the agencies opted to rely on household surveys of willingness to pay (WTP) for wetlands in the benefit-cost analysis.

The WTP-derived benefits, which primarily came from flood protection values, were then allocated to nonadjacent or "isolated" wetlands. Such wetlands were estimated to comprise over half of all wetlands and over 20 percent of total wetland area (Tiner 2003; Cohen et al. 2016). The EPA and Army Corps report concluded that wetland benefits vary by region, ranging from \$26,000 to \$287,000 per acre. A concurrent scientific review supported the 2015 rule's treatment of isolated wetlands, finding that such wetlands generally are "physically, chemically, and biologically integrated with rivers" and that they provide many downstream benefits including storage of floodwater (EPA 2015, p. ES-2)

In contrast, the 2020 NWPR rejects the "significant nexus" criteria and draws on a new EPA and Army Corps impact assessment that largely excludes wetland benefits, citing a lack of credible recent research (EPA 2018; Howard and Shrader 2019).⁵ Since wetlands comprised the largest benefit category in the 2015 assessment, the costs of the 2015 rule were determined to significantly outweigh the benefits. EPA's Science Advisory Board opposed the new rule, stating it "lacks a scientific justification" in relation to the exclusion of wetlands without a direct surface connection to navigable water (Science Advisory Board 2020). Boyle, Kotchen, and Smith (2017) discuss the implications of this agency-level inconsistency in their approach to benefit-cost analysis. Thus, a rigorous analysis of the value of wetlands at a nation-wide scale is needed to inform this policy debate.

C. NFIP

The NFIP was created in 1968 by the federal government to provide affordable flood insurance to property owners. Currently, the program is the dominant flood

³ Available: https://oag.ca.gov/system/files/attachments/press-docs/WOTUS\%20Complaint\%20Filed_05012020.pdf.

⁴ While the term "isolated" lacks a consistent definition in policy documents, the scientific literature defines "geographically isolated wetlands" as wetlands that are surrounded by uplands.

⁵ Excerpt from the 2018 report (EPA 2018): "Many components of the 2015 analysis do not satisfy these requirements. No national level studies concerning WTP for the expansion or preservation of wetland acreage are currently available for the United States, and the US freshwater (noncoastal) wetlands valuation literature is relatively thin. While there are several wetlands valuation studies in the literature, many are context-dependent and not suitable or appropriate for transfer in this analysis."

insurer in the US, with more than 5.1 million policies covering more than \$1.3 trillion of property as of 2018 (FEMA 2020b). To simplify the rate-setting process, flood insurance premiums are based on average historical losses in stratified flood risk zones. The program has come under criticism for failing to charge actuarially fair rates (GAO 2017). More than 20 percent of the policies are heavily subsidized, charging less than one-half of full risk levels (CBO 2014; Council et al. 2015). It is also debatable whether the remaining “full risk” policies are actuarially priced since they do not include a “loading charge” to build up reserves for especially costly years—which was made apparent with Hurricane Katrina in 2005 when the NFIP had to borrow to pay its \$13.2 billion in claims (Michel-Kerjan 2010). Currently, the NFIP owes \$20.5 billion to the US Treasury (Congressional Research Service 2021).

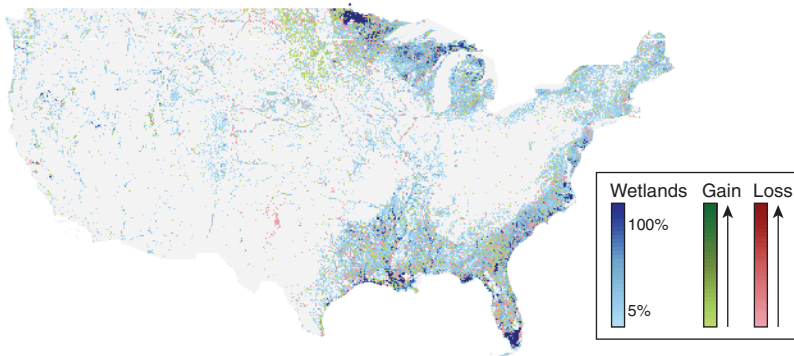
The structure of the NFIP has two important implications for wetland regulation. First, since flood risk is subsidized by the federal government, the financial incentive of private landowners to conserve flood-mitigating wetlands is not aligned with that of society. Second, if reductions in wetland area cause increases in NFIP claims, any change to wetland regulation that removes barriers to developing wetlands could represent a transfer from taxpayers to property owners and developers.

II. Data

We derive our data on the spatial extent of wetlands from the National Land Cover Database (NLCD). The NLCD is a remotely sensed product that provides nationwide data on land cover at 30-meter resolution from 2001 to 2016 (Homer et al. 2020). The latest generation of NLCD, released in 2019, is harmonized across years so that individual years can be compared, facilitating land cover change detection. The NLCD indicates the spatial extent of wetlands was approximately 47.1 million hectares (5.8 percent of the conterminous US) in both 2001 and 2016. Over the 15-year study period, the nation lost approximately 330,000 hectares of wetland area, but these losses were offset by a comparable amount of wetlands gains—thus achieving at a national level the long-standing federal objective of “no net loss” of wetlands. Figure 1 maps the data on wetland extent from the NLCD. Panel A shows the percentage of area covered by wetlands (blue scale). Areas with gains and losses in wetlands over the 2001 to 2016 study period are indicated in green and red, respectively. These changes constitute the primary source of variation used in our identification strategy.

We complement the data on wetland extent with geospatial information on the US water drainage network from the National Hydrography Dataset (NHD) (US Geological Survey 2021), which is used by the EPA and the Army Corps for CWA jurisdictional determinations. We employ the NHD in two distinct ways. First, we classify wetlands as upstream or downstream of each zip code in our sample using the flow direction in NHD’s Watershed Boundary Dataset, enabling our upstream versus downstream difference-in-difference analysis. Second, we compute the distance of all wetland areas in our sample from the NHD surface water network, enabling our evaluation of the relative contribution of “isolated” wetlands to flood mitigation. Panel B of Figure 1 maps the area of wetlands by distance from the surface water network. Wetlands shown in blue represent those

Panel A



Panel B

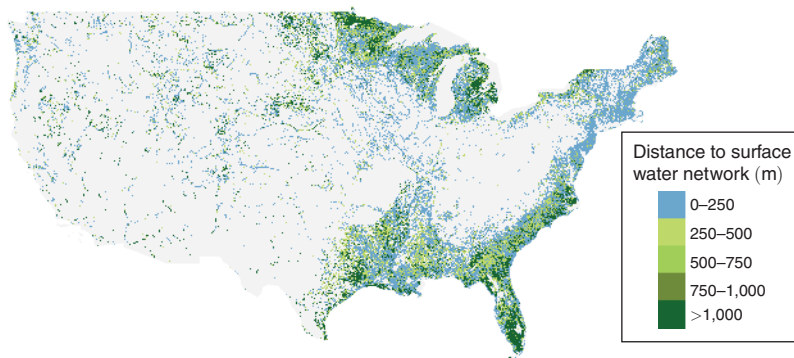


FIGURE 1. WETLANDS IN THE UNITED STATES

Notes: Panel A shows percentage of area covered by wetlands (blue scale). Areas with increases and decreases in wetlands over the 2001 to 2016 study period are indicated in green and red, respectively. Panel B maps the most prevalent type of wetland, separated into riparian (blue) and isolated (green) wetlands. Since no formal definition for isolated wetlands exists, we show isolated wetlands using multiple distance cutoffs. Grid cells are classified as containing wetlands if at least 2.5 percent of the 3 kilometer (km) $\times$ 3 km area is classified as wetland. Data are from the NLCD for the years 2001 and 2016.

within 250 meters of the nearest stream or river. Green areas represent wetlands that are further removed from the surface water network, with the darker shades representing larger distances.

Data on our dependent variable, flood insurance claims, come from the FEMA's NFIP Redacted Claims Dataset (FEMA 2020a). This dataset comprises the NFIP's full claim history and represents more than 2 million transactions. Due to privacy concerns, the NFIP does not provide address-level data; we identify the location of each claims transaction by the property zip code. To control for differences in flood insurance uptake, we also obtain zip-code level data on population, income, number of housing units, and median home values from the US Census (US Census Bureau 2000) and American Community Survey (US Census Bureau 2016a), changes in developed land from the NLCD (area categorized as either low, medium, or high intensity developed where impervious surfaces account for over 20 percent of cover), and local governance as measured by participation in the NFIP's Community Rating System.

To construct our dataset, we aggregate the measures of wetland area from the NLCD to the zip code-level for the years 2001, 2006, 2011, and 2016 (US Census Bureau 2010). We then average NFIP claims over the five-year periods surrounding these dates (i.e., 1999 to 2003 for 2001) and merge the two data sources. We elect to use five-year averages because the amount in NFIP loss dollars paid is highly variable across individual years due to the infrequent nature of flood events.⁶

III. Empirical Strategy

The goal of our empirical estimation is to capture the causal effect of wetlands on flood insurance claims. The primary challenge is that wetland extent is correlated with other factors that drive flood damages. For example, communities with wetter climates and more frequent flooding tend to have more wetlands. Indeed, we find a positive correlation between wetland extent and flood damage. This relationship should not be interpreted as wetlands causing flooding since confounding factors (including precipitation) could be driving the correlation. Similarly, flood damages and real estate development are positively correlated, further raising concerns about omitted variables.

To address identification challenges, we employ three empirical approaches that rely on different assumptions and identifying variation to establish a causal effect. The first is long differences that captures the long-term effect of wetland change on flood outcomes, inclusive of adaptation. The second is a panel analysis that captures shorter-term local effects. The third is an upstream-downstream differences-in-differences, which leverages exogenous geographic variation based on the flow of water to address concerns about endogeneity and unobservable confounders (Duflo and Pande 2007).

The first two approaches are rather standard in economics, so here we elaborate on the motivation behind the upstream-downstream differences-in-differences. This approach takes advantage of the fact that flood risk should be influenced by changes in upstream wetland area but not changes in downstream wetland area (in the case noncoastal areas). Figure 2 illustrates this dynamic. Therefore, we can use changes in downstream wetland area as a natural counterfactual for changes in upstream wetland area; that is, we use downstream changes in wetlands to effectively control for time-varying factors unrelated to flooding that drive both changes in overall wetland extent and changes in NFIP claims. Consider real estate development, for example: wetlands are lost to urban expansion in places with high population growth, which in turn experience higher flood damages due to the larger housing stock. As a result, wetland loss could be misidentified as causing NFIP claims. The upstream-downstream approach addresses this concern under the assumption that real estate development is not systematically biased toward either upstream or downstream areas relative to a given zip code, which we test using the “developed” land cover class from the NLCD and find no significant difference between upstream and downstream development.⁷

⁶We show the sensitivity of our results to using alternative time windows in online Appendix B.

⁷The difference in means for change the proportion of area developed between 2001 and 2016 (upstream less downstream) is -0.0003 (95 percent confidence interval $= -0.0017$ to 0.0012).



FIGURE 2. WETLANDS AND FLOODING SCHEMATIC

Notes: Left: city under nonflood conditions with both upstream and downstream wetlands. Center: city under flood conditions with both upstream and downstream wetlands, in which case the upstream wetlands mitigate flooding, while downstream wetlands do not. Right: city under flood conditions with only downstream wetlands, in which case the wetlands do not mitigate flooding.

Source: Illustration commissioned by authors and created by Kaitlyn Kraybill-Voth.

The upstream-downstream approach also provides insight into flood reduction mechanisms. Wetland loss is associated with two outcomes: (i) the loss of the wetland's sponge-like capacity to absorb and hold floodwaters during rainfall events, and (ii) real estate development along transitional terrestrial-aquatic zones that results in more capital being exposed to flooding, and by extension, higher potential damages (Kousky et al. 2013). The former, which is the “direct protective service” of wetlands, can be isolated in the difference-in-difference model by looking at the effect of upstream wetlands after separately controlling for changes in wetlands locally (i.e., development occurring on former wetlands).

Panel A of Figure 3 illustrates the different sources of variation used in the three identification strategies to establish a causal effect. We use Alabama as an example to plot the variation used by each estimator, with green indicating zip codes with wetland gain and red indicating units with wetland loss. The long difference (LD) estimator exploits the change in wetland area over the 15-year period. The panel fixed effects (Panel) estimator relies on five-year fluctuations in wetland extent.⁸ The difference-in-difference (DID) estimator compares the differential effect of a change in wetland upstream versus downstream. The three approaches still allow for the retrieval of the same empirical object: an estimate of the within-zip code effect of local wetlands on local flood mitigation. We provide further details on the empirical specification of each approach in Section IV.

A. Endogeneity Concerns about Insurance Uptake

Valid endogeneity concerns may remain if flood insurance uptake is correlated with wetland changes. In that case, the estimated effect of wetland changes on flood insurance claims would capture both increased damages to the already insured and changes in NFIP uptake in response to the wetland losses. Ideally, our empirical specification

⁸For the panel model illustration, we plot the 2016 deviations from the average of each zip code as an example.

would control for NFIP policies-in-force. However, information on the number on NFIP policies is not available at the zip code level (our unit of analysis) before 2009.

We address the concern that uptake of flood insurance may be endogenous in several ways: first we select a number of controls in consultation with the existing literature on empirical determinants of NFIP uptake (Kriesel and Landry 2004; Kousky and Michel-Kerjan 2017; Wagner 2019). In all models, we control for changes in population, income, number of housing units, and housing value. We also control for local governance, as measured by participation in the NFIP's Community Rating System, which reflects a community's attitude toward flood risk. Further we flexibly control for changes in built-up land to account for the fact that urban expansion is correlated with both wetland loss and increased insurance uptake as people and capital move into flood plains.

Second, we estimate wetland benefits using county-level data, where information on policies-in-force is available over the full sample period. We show that controlling flexibly for the number of policies does not affect county-level estimates of wetland benefits, giving us confidence that NFIP uptake is not driving our results.

Third, we run our zip code-level analysis with claims per policy as the outcome variable after extrapolating the number of zip code-level policies back to 2001 using a ridge regression prediction model trained on county-level data that leverages the full suite of covariates discussed above (details in online Appendix B).

Finally, we contend that much of the uptake concern is addressed through the upstream-downstream analysis under the assumption that changes in insurance uptake are not related to the relative position of wetlands in a watershed at a small spatial scale like a zip code.

B. Choice of Welfare Measure

We elect to use NFIP claims as our primary outcome variable because they provide a direct measure of actual damages from flooding and likely comprise a significant component of the welfare costs from wetland loss. Additionally, the richness of the data in terms of temporal span, spatial extent and granularity, and consistency in the measurement makes our analysis empirically tractable. However, there are some notable limitations to using NFIP claims: they neither capture flood damages that occur outside the NFIP nor the nonflood mitigation value of wetlands. These factors likely lead to *underestimates* of the value of wetlands.

An alternative approach to using NFIP claims would be hedonic analysis. However, we believe hedonic analysis is ill-suited to the estimation of wetland benefits given that the classic hedonic model assumes a fixed housing stock. Because we are studying the conversion of wetlands into other land uses, including developed area, it is likely that our treatment is related to changes in the supply and composition of the housing stock. Nevertheless, for completeness, we also perform a hedonic analysis and find only limited effects (see online Appendix C). This is expected given that heavy subsidies on flood insurance and uncertainty about wetland benefits likely prevent the full capitalization of wetland value in home prices. The weak results are also in line with Keiser and Shapiro (2019) who find little impact of CWA-driven water quality improvements on housing prices, which they attribute to incomplete information on water pollution and its welfare implications.

Addressing this lack of information is the core motivation for this paper: the EPA and Army Corps cite a lack of credible estimates of the value of wetlands to justify the exclusion of most wetland benefits under the 2020 NWPR rule (EPA 2018; Howard and Shrader 2019).

IV. Models

A. Long Differences

We utilize a long differences approach to model changes in flood insurance claims over time as a function of changes in wetland area, accounting for time-invariant unobservables at the local level and time-trending unobservables at the state-level. We estimate the long differences model:

$$(1) \quad \Delta F_{is} = \beta \Delta W_{is} + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{is},$$

where ΔF_{is} is the change in NFIP claims in zip code i and state s between the years 2001 and 2016. The treatment variable ΔW_{is} represents the change in the spatial extent of wetlands in zip code i over the same period. The vector of covariates, $\mathbf{X}$, includes those described above: changes in population, income, the number of housing units, average housing value, local governance (as measured by participation in the NFIP's Community Rating System) and built-up land. Finally, we include state fixed effects, α_s , to control for any unobserved state-level trends. We cluster standard errors at the county level to control for correlation in the error term over space.

We also estimate an alternative specification that allows flood damages to respond differently to gains and losses in wetlands. This model is identical to equation (1) except that we model changes in NFIP claims as a simple piecewise linear function of changes in wetland area:

$$(2) \quad \Delta F_{is} = \beta_1 \Delta W_{is}^{GAIN} + \beta_2 \Delta W_{is}^{LOSS} + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{is},$$

where W_{is}^{GAIN} describes an increase in the spatial extent of wetlands in zip code i over the period 2001 to 2016 and W_{is}^{LOSS} describes a decrease in wetland area; that is, $W_{is}^{GAIN} = \Delta W_{is}$ for localities with $\Delta W_{is} \geq 0$, and $W_{is}^{LOSS} = -\Delta W_{is}$ for localities with $\Delta W_{is} < 0$.

B. Panel Analysis

We alternatively estimate a panel fixed effects model that relies on shorter-term changes in wetland area to identify the effect of wetlands on flood damages. Specifically, we estimate the model

$$(3) \quad F_{ist} = \beta W_{ist} + \theta \mathbf{X}_{ist} + \alpha_i + \delta_{st} + \epsilon_{ist},$$

where all variables are defined as in equation (1) and the panel contains four observations per zip code corresponding to the four years with wetland measures available from the NLCD: 2001, 2006, 2011, and 2016. The identification relies on

five-year deviations from average wetland area to identify the coefficient of interest. This approach controls for unobservable heterogeneity at the zip code-level as well as state-level time trends in NFIP flood claims. We include the same set of time-varying controls as in equation (1). Standard errors are clustered at the county level to control for correlation in the error term over time and space.

The panel approach allows us to include all years in the NLCD in our analysis; however, this comes at the cost of potentially introducing more measurement error into our data. The NLCD is a remotely sensed product, and although the dataset is designed to be comparable across years, any misclassification of wetlands in a given year will appear as land use change in our data. One general concern with fixed effects estimates is attenuation bias caused by measurement error in the treatment variables. Further, given the sticky nature of land use, it is improbable that a pixel will categorically change its land use in a 5-year period, and even less likely that it will change its land use more than once over the 15-year period. Given that longer-term variation better describes the land use change process, we utilize the panel estimates primarily as a check on the consistency and robustness of our results to using alternative identifying variation.

C. Upstream-Downstream Difference-in-Differences

We employ an upstream-downstream difference-in-difference framework to estimate the effect of wetlands on flood damages using the equation:

$$(4) \quad \Delta F_{is} = \beta \Delta W_{is} + \gamma \Delta W_{is}^{UP} + \lambda \Delta W_{is}^{ALL} + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{is},$$

where ΔW_{is} denotes changes in wetland area within zip code i , ΔW_{is}^{UP} denotes changes in wetland area upstream of zip code i and ΔW_{is}^{ALL} denotes changes in wetland area both upstream and downstream of the zip code. The spatial extent of upstream versus downstream wetland area was computed using NHD flow direction over the full geographic extent of the watershed, but excluding wetlands in the same zip code (see online Appendix A for details). We drop coastal zip codes where downstream wetlands may mitigate flooding in the context of storm surges and cyclone impacts. There are two coefficients of interest: β represents the effect of “local” wetlands (within the same zip code) and is directly comparable to the parameters we estimate using the long differences and panel approaches; γ represents the differential effect of upstream wetlands, which reflect the “direct protective services” channel of flood mitigation. Once again, standard errors are clustered at the county level.

V. Results

A. Main Results

We first establish that wetland loss significantly increases property damages from flooding. Table 1 shows the estimated effect of changes in wetland area on NFIP claims using the three different identification strategies. Columns 1–3 specify the response as linear as in equation (1), while columns 4–6 allow for differential responses to gains

versus losses in wetland area as in equation (2). Our results suggest that flood damages respond differently to gains versus losses in wetland area, indicating that the piecewise specification better describes the shape of the response function.

Across all specifications, we find that wetland loss significantly increases NFIP claims. We estimate that one hectare of “local” (within-zip code) wetland loss is associated with an increase in zip code-level NFIP claims ranging from \$451 under the upstream-downstream difference-in-difference approach (column 5) to \$495 using the long differences approach (column 4). These results are robust to specifying the outcome variable as claims per policy, flexibly controlling for changes in development and precipitation, limiting our results to locations that experienced flooding over the study period, using alternative time windows to calculate flood damages, withholding regional blocks of data, using real instead of nominal values, and aggregating to the county level rather than zip code (online Appendix B). Interestingly, we do not find compelling evidence that wetland gains are associated with decreased flood insurance claims. The estimated effect of wetland gain is small in magnitude and never significantly different from zero.

Panel A of Figure 3 illustrates the different identifying variation used in each empirical approach, as described in the empirical strategy section. Panel B plots the resulting point estimates from the piecewise linear regressions for the local wetland effects. The estimated response functions overlap despite relying on different assumptions and sources of identifying variation.

A key benefit of the upstream-downstream difference-in-differences is that it allows us to separately estimate the effect of upstream wetlands on NFIP claims. We find that one hectare of upstream wetland loss is associated with a \$810 increase in zip code-level NFIP claims. The larger effect size of upstream wetlands is expected given that our measure of local wetlands encompasses wetlands both upstream and downstream of properties in the same zip code—and downstream wetlands should not have any flood mitigation value.

Isolating the effect of upstream wetlands on NFIP claims allows us to disentangle the mechanism through which wetland loss increases flood damages. As discussed above, the conversion of wetlands into developed areas could increase flood damages in two ways: (i) by eliminating the protective ecosystem services wetlands provide against flooding or (ii) by increasing capital exposure in flood-prone areas that were formerly wetlands. The difference-in-difference design breaks this link by estimating the effect of upstream wetlands while controlling for wetland changes in the zip code in which NFIP claims are observed. While it is not clear which mechanism drives the effect of wetland loss within a zip code, the effect of upstream wetlands can be attributed to protective services alone. To our knowledge, these results provide the first empirical evidence of the direct protective services mechanism.

B. Spatial Extensions

We next explore how the flood-reducing benefits of wetlands depend on their location. This is motivated by the fact that the cumulative influence of individual wetlands within watersheds can strongly affect the spatial scale, magnitude, frequency, and duration of water flows (EPA 2015). First, we test for the presence of positive spatial externalities (i.e., whether wetlands produce measurable off-site

TABLE 1—THE EFFECT OF WETLANDS ON FLOOD DAMAGES

	Zip code-level NFIP claims (US\$)					
	LD (1)	DID (2)	Panel (3)	LD (4)	DID (5)	Panel (6)
<i>Wetland effects</i>						
Local wetland change (hectares, or ha)	−229.2 (127.7)	−157.8 (102.1)	−180.9 (83.6)			
Local wetland gain (ha)				−24.1 (116.4)	39.0 (74.7)	153.6 (220.9)
Local wetland loss (ha)				−495.3 (250.8)	−450.8 (247.2)	−461.7 (272.4)
Upstream wetland change (ha)		−500.0 (211.8)				
Upstream wetland gain (ha)					−71.9 (77.6)	
Upstream wetland loss (ha)					−810.4 (342.0)	
<i>Controls</i>						
Developed area (ha)	390.7 (172.0)	337.5 (166.7)	2,863.7 (2,167.9)	372.0 (170.1)	312.0 (165.7)	2,866.5 (2,170.0)
Median income (US\$)	−0.5 (1.1)	−0.6 (1.2)	1.0 (2.2)	−0.5 (1.1)	−0.5 (1.2)	1.0 (2.2)
Population	−6.9 (12.1)	−5.6 (11.4)	−165.6 (167.1)	−6.4 (12.0)	−5.0 (11.4)	−165.3 (167.0)
Number of housing units	77.5 (25.5)	76.3 (29.1)	339.1 (204.5)	76.8 (25.4)	75.7 (28.9)	337.8 (204.5)
Median home value (US\$)	0.2 (0.2)	0.2 (0.3)	0.6 (0.5)	0.3 (0.2)	0.2 (0.3)	0.6 (0.5)
CRS discount (percent)	159,307 (100,644)	169,618 (117,832)	135,364 (111,952)	159,242 (100,657)	168,208 (117,592)	135,424 (111,940)
Watershed wetland change (ha)		−21.0 (15.3)				
Watershed wetland gain (ha)					−48.5 (20.7)	
Watershed wetland loss (ha)					−1.1 (23.8)	
Fixed effects	State	State	Zip, Year	State	State	Zip, Year
Observations	25,734	24,475	93,111	25,734	24,475	93,111

Notes: We estimate the effect of wetlands on NFIP claims at the zip code level using three different identification strategies: long differences (LD), upstream-downstream difference-in-differences (DID), and panel fixed-effects (Panel). We specify the response of flood damages to changes in wetland area as linear in columns (1–3) and allow for differential responses to gains and losses in wetland area in columns (4–6). The DID drops coastal zip codes. Local wetland changes are defined as changes in the spatial extent of wetlands within zip code i (W_i in equation (4)). Upstream wetland changes are changes in the spatial extent of wetlands in all areas upstream of zip code i (W_i^{UP} in equation (4)). Watershed changes are the total change in the spatial extent of wetlands both upstream and downstream of zip code i (W_i^{ALL} in equation (4)). Covariates include developed area, median income, population, number of housing units, median home value, and the mean Community Rating System (CRS) discount rate. Standard errors are clustered by county.

benefits). Then we evaluate how the location of wetlands with respect to the surface water network impacts their value for flood mitigation—a highly relevant matter for WOTUS rulemaking and the CWA.

We utilize the long differences approach in order to obtain more precise and tractable estimates in the spatial and heterogeneity analyses that follow. These models

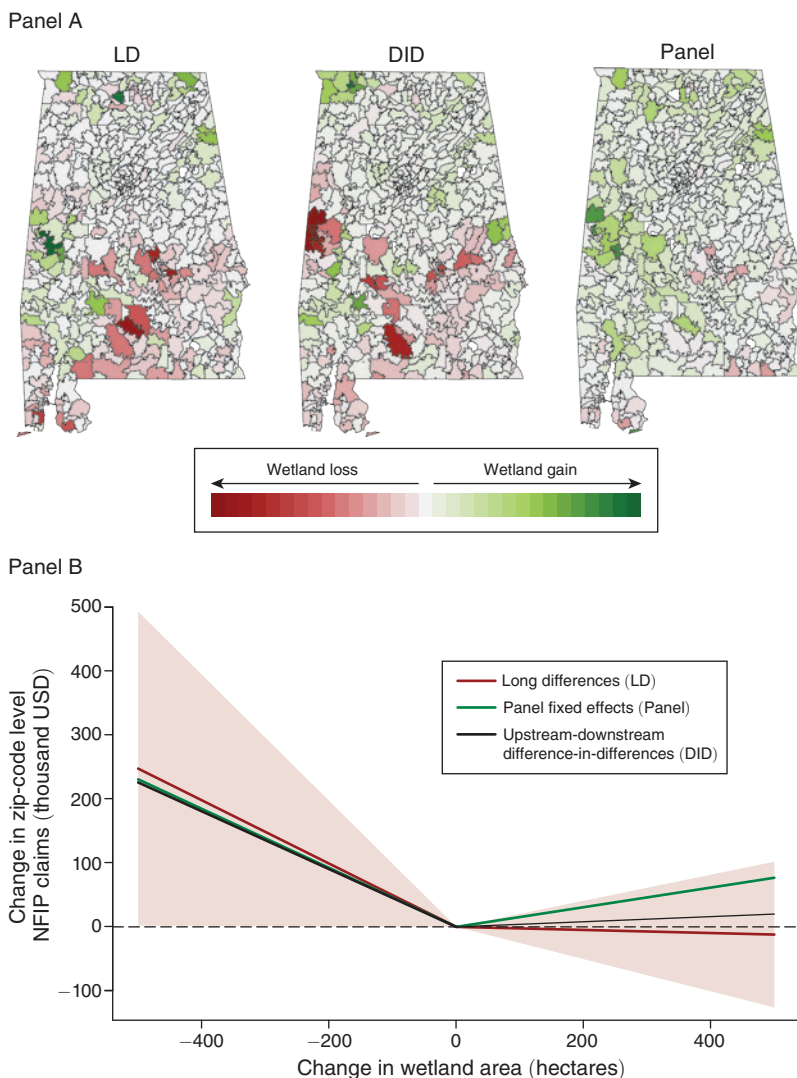


FIGURE 3. ESTIMATES OF THE EFFECT OF WETLANDS ON NFIP CLAIMS USING DIFFERENT IDENTIFICATION STRATEGIES

Notes: Panel A shows the variation used by each estimator in Alabama. Green indicates zip codes with wetland gain over the period 2001 to 2016, and red indicates zip codes with wetland loss. The long difference (LD) estimator exploits the change in wetland area over the 15-year period. The panel fixed effects (Panel) estimator relies on short-term deviations from average wetland changes to identify coefficients. We plot these deviations for the year 2016. The difference-in-difference (DID) estimator compares the differential effect of a change in wetland upstream versus downstream. We illustrate this concept by plotting upstream change in wetland area less downstream change in wetland area. Panel B compares the estimated effect of wetlands on NFIP claims using the LD, Panel, and DID estimators at the national level. Shaded polygons show the 95 percent confidence interval for the LD estimates.

involve estimating additional parameters to decompose wetland impacts by their location and characteristics. Given that the difference-in-difference model separately estimates the effect of upstream, local, and watershed-level wetland change as part of its design, adding in additional interactions yields less precise results that can be difficult to interpret. Further, the difference-in-difference model omits

coastal areas where downstream wetlands may mitigate flooding in the context of storm surges and cyclone impacts. But for added assurance, we utilize both the long differences model and the upstream-downstream difference-in-difference results to estimate the overall value of wetlands with spatial spillovers based on the sum of local and upstream (see Section VB) and find very similar results.

Spatial Lag Model.—We test for the presence of spatial spillovers using a spatial lag model (Cressie and Wikle 2011), where zip code i 's damages from flooding are affected by i 's wetland extent, plus the wetland extent of all *upstream* neighbors j whose centroids fall within concentric rings around i 's centroid with 10 km widths (see Figure 4 panel A). The estimating equation is

$$(5) \quad \Delta F_{is} = \sum_{d=1}^6 (\beta_{d1} W_{dis}^{GAIN} + \beta_{d2} W_{dis}^{LOSS}) + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{dis},$$

where W_{dis}^{GAIN} and W_{dis}^{LOSS} denote the wetland gain and loss, respectively, in distance band d (e.g., 10 to 20 km away from zip code i). The parameters of interest are the β_{d2} , which represent the change in zip code i 's NFIP claims caused by a one hectare decrease in the spatial extent of wetlands that are located a given distance (e.g., 10 to 20 km) away from zip code i .

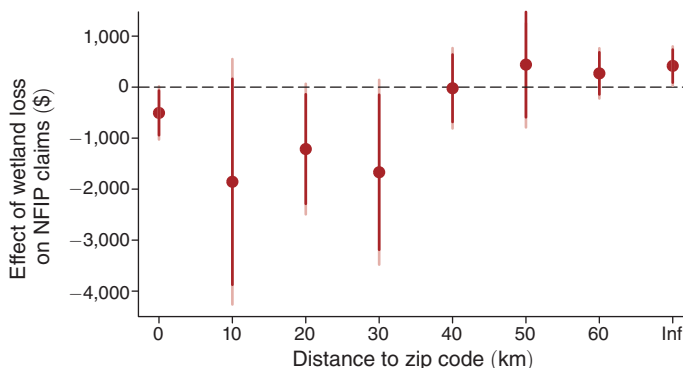
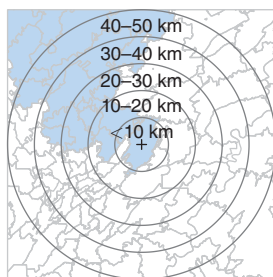
Panel A of Figure 4 plots the spatial lags estimated using this model. Despite some statistical uncertainty, our results suggest that there are spatial spillovers in the benefits of wetlands such that wetlands have a substantially larger value to society than that which accrues to local property owners. Indeed, the flood mitigation value of wetlands to local property owners (i.e., in the same zip code) amounts to less than 30 percent of their benefits to all downstream users.

Wetland Valuation with Spillovers.—We next estimate the overall value of wetlands after accounting for spatial spillovers. Using the results of our spatial lag model based on equation (5), we calculate the total estimated damages resulting from the average hectare of wetland loss in our sample with the following equation:

$$(6) \quad \text{Average value} = \frac{\sum_i \sum_d (\beta_{d2} \times W_{di}^{LOSS})}{\sum_i W_i}.$$

Here the numerator represents the total cost of wetland loss in the United States between 2001 and 2016. It is computed by (i) estimating zip code-level costs of wetland loss by multiplying the coefficient estimates for wetland loss at each distance (β_{d2}) by the amount of wetland loss in the corresponding distance band d , and summing across distance bands, and then (ii) summing over zip codes. The denominator represents the total area of wetland loss in the United States between over the study period. The interpretation of this average value is thus the average loss in flood mitigation value, measured in dollars per year, resulting from one hectare of wetland loss in the United States between 2001 and 2016. We calculate confidence intervals by bootstrapping this estimation procedure (both equations (5) and (6)) with 1,000 iterations. We find that each hectare of wetland loss increases NFIP claims by \$1,897 (95 percent confidence interval = \$561 to \$3,233).

Panel A. Spatial lag model



Panel B. Distance from surface water network

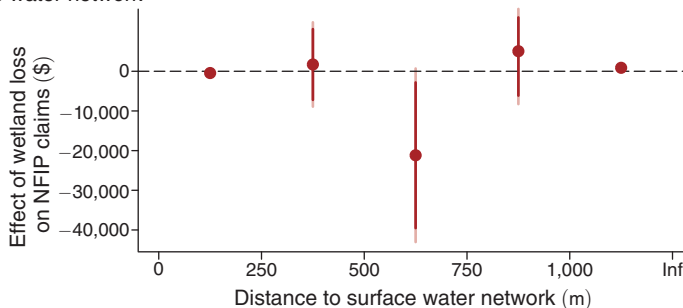
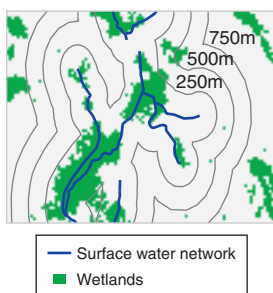


FIGURE 4. SPATIAL ANALYSES OF THE VALUE OF WETLANDS FOR FLOOD MITIGATION

Notes: Panel A: spatial lag model. Left shows an example of the annuli used to construct spatial lags, with upstream zip codes highlighted in blue. Right plots the coefficients on the spatial lags for the effect of wetland loss on zip code level NFIP claims. Panel B: Analysis of how distance from the surface water network affects the flood mitigation benefits of wetlands. Left shows an example of how the distance from the surface water network is calculated. Right plots the estimated effect for each distance bin. Lines show 90 percent (red) and 95 percent (pink) confidence intervals.

We also estimate the value of wetlands with spillovers using the upstream-downstream difference-in-difference design. We follow the same general procedure as in equation (6) but instead of summing over distance bands, we calculate the total cost of wetland loss as the sum of local and upstream losses. Using this alternative approach, we arrive at nearly identical average cost of \$1,843 (95 percent confidence interval = \$624 to \$3,062) per hectare of wetland loss.

However, this average value masks significant heterogeneity among wetlands. For example, if we calculate the same value for wetlands located in developed areas (those with over 10 percent built-up area), we find that one hectare of wetland loss between 2001 and 2016 costs society \$8,294 (95 percent confidence interval = 1,232 to 15,355) in annual flood mitigation value.⁹ We explore other important dimensions of heterogeneity in Section VC.

⁹This value is estimated by adding an interaction term between wetland area changes and a binary indicator for whether the zip code is at least 10 percent built-up area, as measured by the NLCD. Approximately 25 percent of zip codes are considered developed using this threshold.

Distance to Surface Water.—Next we evaluate how wetland location with respect to the surface water network affects flood mitigation value. This analysis is motivated by the recent NWPR rule change to the definition of WOTUS that eliminates federal protections under the CWA for “isolated” wetlands (i.e., those without a surface water connection to streams or rivers). We first calculate how far each wetland area is from the nearest surface water connection using the detailed mapping of the US water drainage network contained in the NHD. We then bin these distances in increments of 250 meters and jointly estimate the effect of wetlands in each distance bin using the equation

$$(7) \quad \Delta F_{is} = \sum_k^5 (\beta_{1k} W_{kis}^{GAIN} + \beta_{2k} W_{kis}^{LOSS}) + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{is},$$

where k denotes the discrete distance bins (e.g., 0 to 250 meters from the nearest surface water connection). All other variables are defined as for the long difference in equation (2).

Panel C of Figure 4 displays the results. We find that wetlands located 500 to 750 meters from the nearest stream or river provide economically large and statistically significant flood mitigation benefits (\$21,178 per hectare). We do not find evidence of flood mitigation benefits for wetlands located less than 500 meters or greater than 750 meters from the nearest surface water connection. This same pattern holds for wetlands in developed areas, albeit with even larger benefits for wetlands at intermediate distances from the surface water network (\$63,276 per hectare).¹⁰

Our findings are consistent with the hydrological concept that wetlands reduce flood damages by “acting like a sponge.” Intuitively, wetlands that are directly adjacent to a stream or river may already be fully saturated and not have excess capacity to absorb excess floodwater. On the other hand, wetlands closer to the stream network have greater hydrologic connectivity than those further away, all things being equal (EPA 2015; Leibowitz et al. 2018). Thus wetlands far from the nearest surface water connection are less likely to intercept floodwaters before causing damage. Indeed, we find that the most valuable wetlands for flood mitigation are those at an intermediate distance, ones which would have excess capacity to absorb floodwater but still likely to be located between floodwaters and residential properties.

C. Heterogeneity

Regional Heterogeneity.—We now explore regional heterogeneity across the US in the flood mitigating effects of wetlands. It is worth noting the complexity of floods: they occur through several hydrological mechanisms including high tidal levels (i.e., coastal flooding), direct precipitation (i.e., pluvial flooding), high groundwater levels (i.e., groundwater flooding) or high river flows (i.e., fluvial flooding). The impacts of floods are also highly influenced by civil infrastructure like levees and canals. Further, given the variety of wetlands types, which can encompass forestland, grassland, salt marshes, and peat bogs, the hydrology literature is

¹⁰This value is estimated by adding an interaction term between wetland area changes in each distance bin and a binary indicator for whether the zip code is at least 10 percent built-up area, as measured by the NLCD. Approximately 25 percent of zip codes are developed using this threshold. See online Appendix B for details.

often hesitant to generalize flood reduction properties across all wetlands (Acreman and Holden 2013; Bullock and Acreman 2003).

To address these concerns we assess whether the impact of wetlands on flood damages differs by geographic region. Specifically, we examine the differential effect of wetlands on either side of the 100° meridian, a boundary long thought to separate the humid eastern United States and the arid Western plains (Powell et al. 1879). We also evaluate the effect of wetlands in nine broad ecoregions: Eastern Temperate Forests, Northern Forests, Tropical Wet Forests, the Great Plains, North American Deserts, Southern Highlands, Northern Forested Mountains, Marine West Coast, and Mediterranean California (EPA 2020).¹¹ Ecoregions were derived by Omernik (1987) in collaboration with the EPA to highlight areas generally similar in their underlying ecosystems and environmental resources and to serve as a spatial framework for ecosystem research and management (EPA 2020b). A map of ecoregions in the United States can be found in online Appendix Figure A2.

Figure 5 shows our estimates for the effect of wetlands on NFIP claims in each region with the long difference. We find significant regional heterogeneity, with the largest benefits east of the 100° meridian, as well as in the Great Plains, Eastern Temperate Forests, and Tropical Wet Forests. For example, we estimate that a hectare of wetlands east of the 100° meridian reduces NFIP claims by \$4,645, but find no discernible effect west of the 100° meridian where we estimate a precise zero. It is worth noting that 90 percent of US wetlands are located east of the 100° meridian, as seen in Figure 1. Wetlands in the Great Plains have the highest estimated flood mitigation potential, with a value of \$7,268 per hectare, followed by those in Eastern Temperate Forests (\$3,232 per hectare) and Tropical Wet Forests (\$2,768 per hectare). These results are robust to subsetting our sample to only include zip codes with considerable wetland area (greater than ten hectares) or only including zip codes where some flooding occurs, as indicated by positive flood insurance claims over the study period (see online Appendix B).

Heterogeneity by Ultimate Land Use.—The type of land use conversion may influence the consequences of wetland loss. There is likely a differential impact of wetlands converted to an impervious surface (e.g., a parking lot) versus wetlands drained for agriculture given that cropland retains some water-holding capacity. To address this, we categorize all areas that transition from wetland in 2001 to nonwetland in 2016 by the land use class they are replaced by, and vice versa for wetland gain.¹² Excluding open water, we find that development accounted for 35 percent of wetland loss and 0 percent of wetland gain, which matches our intuition that developed land is unlikely to revert back to wetlands. Cropland and pasture together accounted for 35 percent of wetland loss and 58 percent of wetland gain.

¹¹ We combine Southern Semi-Arid Highlands and Temperate Sierras into Southern Highlands due to the small number of observations in each of these neighboring regions. If estimated separately, the estimates are still not significantly different from zero but are less precise.

¹² Over 50 percent of both wetland gains and losses involve a transition either from or to open water. Such a high and consistent proportion for wetland loss and gain can be expected given that wetlands often border surface water bodies and the fact that there is a natural gradient between wetland and water. While wetlands can become open water as a result of sea level rise and land subsidence, it appears that much of the conversion to/from open water is due to natural processes and the classification of these transitional land classes (Homer et al. 2020).

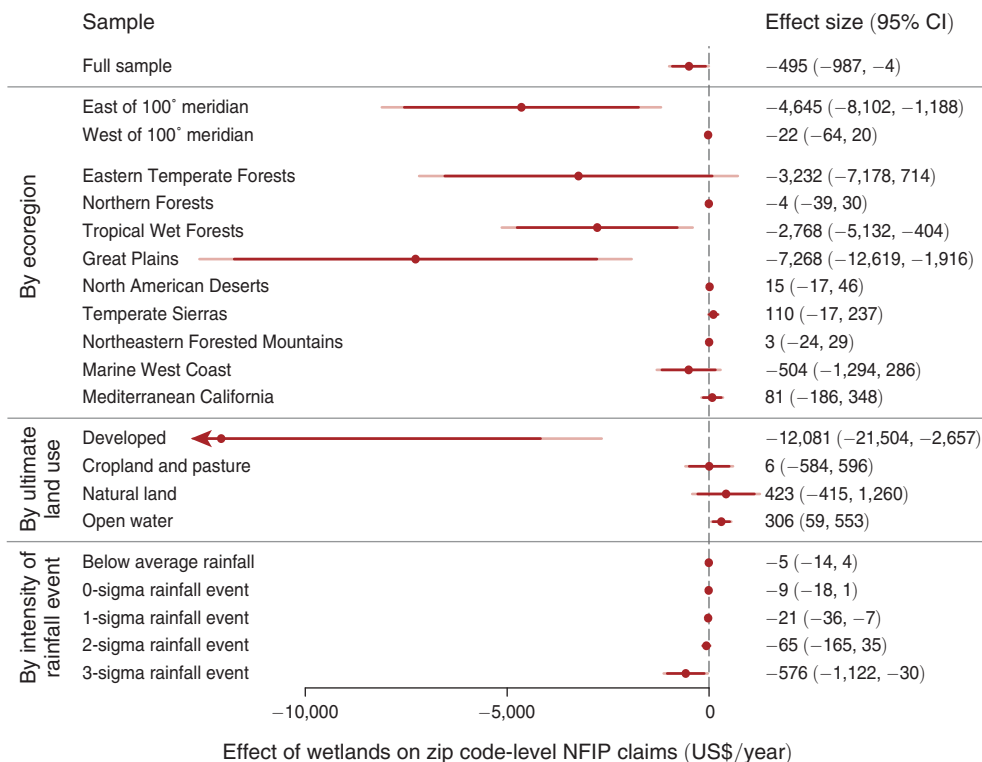


FIGURE 5. HETEROGENEOUS EFFECTS OF WETLAND LOSS ON FLOOD DAMAGES

Notes: Dots are point estimates and lines represent 90 percent (red) and 95 percent (pink) confidence intervals. We show heterogeneity by ecoregion, ultimate land use (the land use class that wetlands were replaced by), and intensity of rainfall events in the month for which NFIP claims are observed. See text for descriptions.

Other natural land uses (forest, grassland, shrubland, open areas, and barren land) accounted for 30 percent of wetland loss and 42 percent of wetland gain.

To test for differential flood mitigation impacts by land use, we run our long difference model separating out area of wetland loss by the land use class the wetlands were replaced by. Results are shown in Figure 5. We find that the loss of wetlands only increases flood damages in the event that wetlands are converted to developed land (defined by the NLCD as areas for which impervious surface covers at least 20 percent of total cover). For each hectare of wetlands converted to developed area, zip code-level NFIP claims increase by \$12,080. We find little detectable effect of conversion to other land uses.

Precipitation and the Effect of Wetland Loss.—Finally, we test whether wetlands have differential impacts on NFIP claims conditional on flood intensity. In other words, we assess whether wetlands are more effective at mitigating floods during low, medium or high levels of rainfall. This is an important question given that extreme precipitation events are expected to become more frequent with climate change (Donat et al. 2016).

We utilize our panel model approach to take advantage of the fact that we can observe both NFIP claims and rainfall events at high temporal frequency. We model monthly zip code-level NFIP claims as a function of wetland area changes, conditional on monthly precipitation using the equation

$$(8) \quad \Delta F_{imt} = f(\Delta W_i^{GAIN}|P_{imt}) + h(\Delta W_i^{LOSS}|P_{imt}) + \theta \Delta \mathbf{X}_i + \alpha_i + \delta_{mt} + \epsilon_{imt},$$

where P_{imt} are monthly precipitation levels (PRISIM Climate Group 2021) and all other variables are defined as in equation (3). The functional forms of $f(\cdot)$ and $h(\cdot)$ exploit linear interactions between wetland area changes and monthly precipitation binned by their standard deviation from the location-specific monthly mean. Specifically, we include four precipitation bins: below average precipitation, 0-sigma precipitation events (precipitation between 0 and 1 standard deviation above the mean), 1-sigma precipitation events, 2-sigma precipitation events, and 3-sigma precipitation events. Because the distribution of monthly rainfall within a location is approximately normal, 1-sigma, 2-sigma, and 3-sigma events have probabilities of 13.6 percent, 2.1 percent, and 0.1 percent, respectively.

Our results are shown in Figure 5. We find that wetland loss only increases NFIP claims in months with 3-sigma precipitation events. In these months, we estimate that one hectare of wetland loss is associated with a \$576 increase in zip code-level NFIP claims. This finding is consistent with the intuition that wetland loss should only affect NFIP claims when flooding occurs (i.e., when there is an extreme precipitation event).

D. Flood Mitigation Benefits Relative to Conservation Costs

Both the flood mitigation value of wetlands and the cost of conserving these natural landscapes depend on local development levels. More developed areas have more exposed capital and thus greater potential for wetlands to reduce flood damages to buildings and other assets. At the same time, land values tend to be higher in more populated areas, increasing the cost of conserving wetlands. Taking these opposing factors into consideration, this section provides guidance on how to target conservation efforts towards high-benefit, low-cost wetlands.

First, we reestimate the effect of wetland area changes on flood damages, this time allowing the response to vary based on local levels of development. To implement this approach, we compute the percent of developed area for each zip code. Then we capture heterogeneous patterns of wetland benefits via the model

$$(9) \quad \Delta F_{is} = g(\Delta W_{is}^{GAIN}|D_{is}) + l(\Delta W_{is}^{LOSS}|D_{is}) + \theta \Delta \mathbf{X}_{is} + \alpha_s + \epsilon_{is},$$

where D_{is} is the sample-period average percent developed area in zip code i and other variables are defined as in equation (2). The functional forms of $g(\cdot)$ and $l(\cdot)$ are modeled as linear interactions between wetland area changes and each quintile of D_{is} . This approach allows us to estimate the differential effect of wetland losses in areas with lower versus higher levels of development. Given the significant regional

heterogeneity in wetland benefits documented in Section VC, we estimate this equation separately for each ecoregion.

We approximate conservation costs using high-resolution maps of the value of private lands (Nolte 2020). This dataset was produced by training an ensemble of machine learning models on 6 million land sales across the contiguous US with the purpose of assessing trade-offs between public and private benefits from land use decisions.

The benefit of wetlands, in terms of avoided flood damages, are estimated in annual flows. The conservation cost is a one-time, upfront investment to purchase the land and ensure it will not be developed. In order to compare the costs and benefits of wetland conservation, we compute the “payback period” for US wetlands. The payback period is simply the number of years in the future at which the expected present value of annual benefits from wetland conservation exceed the initial investment of purchasing the land.¹³

For the median hectare of wetlands in the continental United States, we find that the societal benefits from reduced flooding outweigh the cost of conserving the wetland within six years. However, such a national average masks heterogeneity in the net benefits of wetlands across space. Figure 6 plots spatially resolved estimates of the payback period for all US wetlands. These estimates can be used to identify high-benefit, low-cost areas to prioritize wetland conservation. To facilitate such efforts, we make our gridded estimates of wetland benefits and approximate payback periods publicly available to researchers and decision-makers.¹⁴ We note that these estimates are likely conservative: they do not incorporate the positive spatial spillovers of wetlands to downstream communities or include their nonflood mitigation value (e.g., recreation, wildlife habitat).

VI. Discussion

Flooding is the most costly and frequently occurring natural disaster, and intense rainfall events are expected to increase with climate change (Donat et al. 2016). To our knowledge, this paper is the first to estimate the causal effect of wetlands on flood damages across the continental United States. Our main results suggest that each hectare of wetland loss that occurred between 2001 and 2016 increased NFIP claims by \$1,840 annually when accounting for spatial spillovers. This amount increases to more than \$8,000 for wetland loss in relatively developed areas. Our results are robust across three identification strategies utilizing different sources of temporal and spatial variation. To put our estimates in context, we calculate the amount of NFIP claims attributable to wetland loss over the 2001 to 2016 study period: we estimate that the 331,000 hectares of wetlands lost over this period increased annual NFIP claims by more than \$600 million per year, which amounts to 23 percent of the average annual program-wide NFIP payouts between 2001 and 2016 (\$2.66 billion annually).

In a back-of-the-envelope calculation, we apply our estimate of \$1,840 per hectare to the 47 million hectares of wetlands in the continental US to calculate the

¹³To calculate the net present value of annualized flood mitigation benefits, we use a discount rate of 3 percent.

¹⁴View and download the maps at <https://hannahdruckenmiller.com/code>.

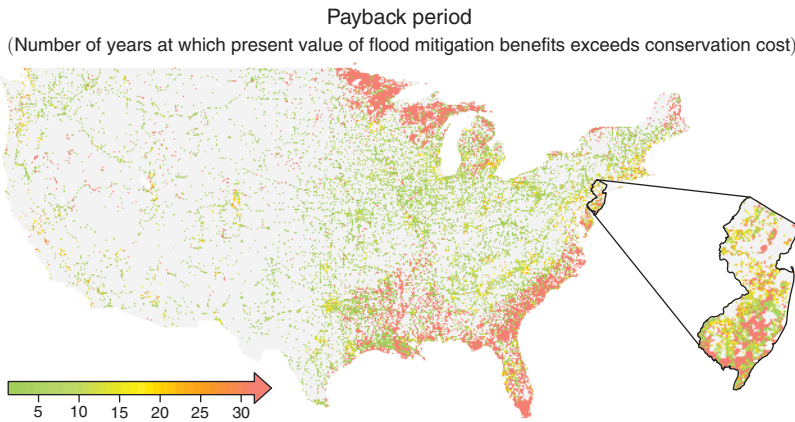


FIGURE 6. SPATIALLY RESOLVED COMPARISON OF WETLAND BENEFITS AND CONSERVATION COSTS

Notes: We plot the payback period for conserving wetland areas, defined as the number of years in the future at which the expected value of annual benefits from wetland conservation exceeds the initial investment in purchasing the land. Wetland benefits mapped here only include flood mitigation value and differ across space according to ecoregion and local levels of development (see equation (9)). The net present value of annualized flood mitigation values is computed using a discount rate of 3 percent. Conservation costs are approximated as the market value of private land (Nolte 2020).

approximate value of US wetlands to society. We estimate that US wetlands provide \$1.2 to \$2.9 trillion, using 7 percent and 3 percent discount rates respectively, in flood mitigation value. While this figure does not reflect the significant spatial heterogeneity in the value of wetlands, it provides an idea of the likely range in the overall value of wetlands for flood mitigation.

Interestingly, we find no discernible effect of increases in wetland area on flood insurance claims, calling into question whether the compensatory mitigation requirements under the CWA are achieving their intended objective of offsetting the adverse impacts of wetland loss. If wetland gains do not offset the reduction in ecosystem services occurring as a result of wetland losses, policy ought to emphasize conservation of existing wetlands. However, we emphasize that our study design does not specifically investigate wetland areas that were created, restored, or enhanced under compensatory mitigation. CWA requirements only account for a portion of overall wetland gain—other drivers include abandonment of tiled/drained agricultural lands, storm surges, land subsistence, and conservation-oriented wetland restoration.

Nevertheless, we believe research into the efficacy of the compensatory mitigation is warranted. On one hand, it is possible that human-made wetlands provide less flood mitigation potential than mature, naturally occurring ecosystems. Conversely, prior research has found that human-made wetlands can have a higher economic value because they are constructed with the specific purpose of providing services for human use (Ghermandi et al. 2010). Net water holding capacity of wetland gain versus wetland loss may also vary: as explained in the heterogeneity analysis by type of land use change, lost wetlands are far more likely to become developed lands, while gained wetlands are almost never from developed land.

This paper also provides the first empirical estimate of the direct protective services provided by wetlands. Converting wetlands into developed areas can increase flood damages in two ways. First, wetlands mitigate flooding by absorbing and holding floodwaters (Acreman and Holden 2013) so converting these “natural sponges” to impervious surfaces increases the severity of flood events. At the same time, real estate development increases capital exposure in flood-prone areas that were formerly wetlands (Kousky et al. 2013). Our upstream-downstream analysis allows us to isolate these effects. We find that the benefits of wetlands are driven mainly by the first pathway involving protective services and that wetland losses cause flood damages by altering hydrological processes.

Additionally, we conduct a number of extensions that exploit the geospatial nature of our data to inform the design of wetlands policy. First, we document spatial spillovers in the flood mitigation benefits of wetlands up to 50 km away. Hydrological connections between wetlands and downstream waters often span long distances: in Prairie Pothole Region, for example, lakes can expand up to 40 km to form a direct surface-water connection with “isolated” wetlands (Vanderhoof and Alexander 2016). Nonfloodplain wetlands can be directly connected to the river network over long distances through subsurface or groundwater flows (EPA 2015; Cohen et al. 2016). Our result provides new empirical evidence for the oft cited theory that regulation may be required to achieve the optimal provision of wetlands due to the presence of positive spatial externalities.

Second, we evaluate how the location of wetlands relative to the surface water network affects their value. We find that the most valuable wetlands for flood mitigation are located 500 to 750 meters from the nearest stream or river. Such relatively isolated wetlands can act as a “sink” for excess water, sediment, and pollutants, preventing their export to downstream waters (EPA 2015). Our result is at odds with the 2020 NWPR’s interpretation of WOTUS that eliminates federal protections for the nearly one-half of wetlands lacking a direct surface water connection. Our results better align with the thresholds of 1,500 feet (460 meters) and 4,000 feet (1,220 meters) referenced in the 2015 WOTUS interpretation regarding varying conditions to determine adjacency to navigable waters.

Our heterogeneity analysis shows how wetland benefits vary by region, type of land use conversion, and the intensity of precipitation events—findings which have scientific or policy insights. For example, we see considerable heterogeneity across US ecoregions. Regionally, we see large flood mitigation benefits in the eastern United States and little to no benefits in the western United States. However, we emphasize that wetlands in the western United States likely provide benefits that are not captured in our specific outcome variable, NFIP flood claims. Nevertheless, the variation across ecoregions suggests that a decentralized implementation of federal policy may be more appropriate for wetland regulation.

Next, when decomposing wetland loss by ultimate land use conversion, we find that wetlands lost to development (as opposed to cropland or another ecosystem type) drive our estimates of flood damages from wetland loss. This is in line with scientific understanding that wetlands mitigate flooding through their water-holding and infiltration capacity as well as their ability to reduce the velocity of floodwaters. Thus the net loss of these functions is greatest when wetlands are converted to an impervious surface. This validates Section 404’s primary focus on the dredging

and filling wetlands for the purposes of development as opposed to prior converted cropland.

Finally, we find the flood mitigation benefits of wetlands to be greatest during anomalously high precipitation events. Since the magnitude and frequency of such extreme events are projected to increase over time, wetlands may play an important role in climate change adaptation strategy vis-à-vis their ability to reduce future flood damages.

There are several important limitations to our study. We estimate the effect of all wetlands on flood damages; however, we recognize that the term “wetlands” describes a diverse set of ecosystems with different levels of flood mitigation potential. Our estimates can be interpreted as the average treatment effect of wetlands, but future research should investigate the contribution of wetland by land cover (e.g., herbaceous versus forested) and type (e.g., swamps, marshes, bogs). Indeed, the regional heterogeneity we see in flood-reducing benefits of wetlands suggests that not all wetlands function in the same way.

Another limitation of our study is that there may be selection into what types of wetlands are converted to other land uses. For example, landowners have a higher incentive to build on wetlands in more developed areas because they tend to have higher property values. Since we find that wetlands in more developed areas have higher flood mitigation potential, this type of selection would suggest that we overestimate the value of the average hectare of wetlands in the United States. On the other hand, selection into wetland loss may be biased in the opposite direction due to the strict regulations surrounding the filling or dredging of wetlands under Section 404 of the CWA. Indeed, since it is more difficult to obtain a permit to develop wetlands that the EPA and Army Corps regard as providing more ecosystem services, development may favor wetlands with lower than average flood mitigation value. This type of selection would suggest that our estimates underestimate the value of the average hectare of wetlands.

Our estimates likely represent a lower bound on the flood mitigation value of wetlands because our dependent variable, NFIP claims, only captures one component of property damages from flooding. While the NFIP is the dominant provider of flood insurance in the US, less than 15 percent of American homeowners participate in the program and the Congressional Budget Office estimates that NFIP claim payments represent just 16 percent of annual flood damages to the residential sector (CBO 2019). It is difficult to extrapolate without knowing the spatial distribution of NFIP policyholders, but if one takes this proportion as given, our estimates would suggest that each hectare of wetland loss increases residential flood damages by approximately \$12,300. Still, this estimate does not capture damages to nonresidential property like commercial and governmental properties, as well as farms and crop yields.

Further, we focus our analysis on flood mitigation alone and do not contemplate the value of wetlands in relation to other ecosystem services. Wetland restoration, for example, limits the amount of nitrogen discharged into the Gulf of Mexico, thereby reducing hypoxia extent and coastal deadzones (Mitsch et al. 2005). As such, it is not surprising that our per hectare valuation falls well below previous EPA estimates of the value of wetlands derived from willingness to pay surveys, which take into account not only their value for flood control, but also for

fishing, hunting, fur trapping, recreation, water filtration, aesthetics and wildlife habitat (EPA 2013).

Nonetheless, as lower bounds, our estimates represent the cost to US taxpayers of converting wetlands to other uses—and thus have policy implications for the highly subsidized NFIP as well as the jurisdictional debates around wetland protection under the CWA, which is one of the most important federal regulations governing land use in the United States and has large impacts on zoning decisions, urban development, and consequently aggregate economic growth (Hsieh and Moretti 2019). As flood events intensify with climate change and development pressures continue, efficient flood mitigation policy that properly accounts for wetland-related public good provision becomes increasingly important.

REFERENCES

- Acreman, M., and J. Holden. 2013. “How Wetlands Affect Floods.” *Wetlands* 33 (5): 773–86.
- Alderman, Katarzyna, Lyle R. Turner, and Shilu Tong. 2012. “Floods and Human Health: A Systematic Review.” *Environment International* 47 (1): 37–47.
- Aon. 2019. *Weather, Climate and Catastrophe Insight: 2019 Annual Report*. London, UK: Aon.
- Barbier, Edward B. 2013. “Valuing Ecosystem Services for Coastal Wetland Protection and Restoration: Progress and Challenges.” *Resources* 2 (3): 213–30.
- Boyle, Kevin J., Matthew J. Kotchen, and V. Kerry Smith. 2017. “Deciphering Dueling Analyses of Clean Water Regulations.” *Science* 358 (6359): 49–50.
- Brander, Luke M., Raymond J. G. M. Florax, and Jan E. Vermaat. 2006. “The Empirics of Wetland Valuation: A Comprehensive Summary and a Meta-Analysis of the Literature.” *Environmental and Resource Economics* 33 (2): 223–50.
- Bullock, Andy, and Mike Acreman. 2003. “The Role of Wetlands in the Hydrological Cycle.” *Hydrology and Earth System Sciences* 7 (3): 358–89.
- Cohen, Matthew J., Irena F. Creed, Laurie Alexander, Nandita B. Basu, Aram J.K. Calhoun, Christopher Craft, Ellen D’Amico et al. 2016. “Do Geographically Isolated Wetlands Influence Landscape Functions?” *Proceedings of the National Academy of Sciences* 113 (8): 1978–86.
- Congressional Budget Office (CBO). 2014. *Understanding FEMA’s Rate-Setting Methods for the National Flood Insurance Program*. Washington, DC: Congressional Budget Office.
- Congressional Budget Office (CBO). 2019. *Expected Costs of Damage from Hurricane Winds and Storm-Related Flooding*. Washington, DC: Congressional Budget Office.
- Congressional Research Service. 2021. *Introduction to the National Flood Insurance Program (NFIP)*. Washington, DC: Congressional Research Service.
- Costanza, Robert, Octavio Pérez-Maqueo, M. Luisa Martinez, Paul Sutton, Sharolyn J. Anderson, and Kenneth Mulder. 2008. “The Value of Coastal Wetlands for Hurricane Protection.” *AMBIO: A Journal of the Human Environment* 37 (4): 241–48.
- Costanza, Robert, Rudolf De Groot, Paul Sutton, Sander Van der Ploeg, Sharolyn J. Anderson, Ida Kubiszewski, Stephen Farber, and R. Kerry Turner. 2014. “Changes in the Global Value of Ecosystem Services.” *Global Environmental Change* 26: 152–58.
- Council, National Research, et al. 2015. *Tying Flood Insurance to Flood Risk for Low-Lying Structures in the Floodplain*. Washington, DC: National Academies Press.
- Cowardin, Lewis M., and Francis C. Golet. 1995. “US Fish and Wildlife Service 1979 Wetland Classification: A Review.” In *Classification and Inventory of the World’s Wetlands*, edited by C. Max Finlayson and A. G. van der Valk, 139–52. New York City, NY: Springer.
- Cressie, Noel, and Christopher K. Wikle. 2011. *Statistics for Spatio-Temporal Data*. Hoboken, NJ: John Wiley & Sons.
- Dahl, Thomas E. 2011. *Status and Trends of Wetlands in the Conterminous United States 2004 to 2009*. Washington, DC: US Fish and Wildlife Service and US Department of the Interior.
- Dewitz, Jon. 2021. “National Land Cover Database (NLCD) Products [Dataset].” <https://doi.org/10.5066/P9KZCM54>.
- Donat, Markus G., Andrew L. Lowry, Lisa V. Alexander, Paul A. O’Gorman, and Nicola Maher. 2016. “More Extreme Precipitation in the World’s Dry and Wet Regions.” *Nature Climate Change* 6 (5): 508–13.

- Doocy, Shannon, Amy Daniels, Sarah Murray, and Thomas D. Kirsch.** 2013. "The Human Impact of Floods: A Historical Review of Events 1980-2009 and Systematic Literature Review." *PLOS Currents* 5 (1). DOI: 10.1371/currents.dis.f4deb457904936b07c09daa98ee8171a.
- Dufflo, Esther, and Rohini Pande.** 2007. "Dams." *Quarterly Journal of Economics* 122 (2): 601–46.
- Environmental Protection Agency (EPA).** 2013. *Economic Analysis of Proposed Revised Definition of Waters of the United States*. https://archive.epa.gov/epa/sites/production/files/2014-03/documents/wus_proposed_rule_economic_analysis.pdf.
- Environmental Protection Agency (EPA).** 2015. *Connectivity of Streams and Wetlands to Downstream Waters: A Review and Synthesis of the Scientific Evidence*. <https://www.federalregister.gov/documents/2015/01/15/2015-00339/connectivity-of-streams-and-wetlands-to-downstream-waters-a-review-and-synthesis-of-the-scientific>.
- Environmental Protection Agency (EPA).** 2018. *Economic Analysis for the Proposed Revised Definition of 'Waters of the United States'*. Washington, DC: Environmental Protection Agency.
- Environmental Protection Agency (EPA).** 2020. "Ecoregions of North America: Level 1 Ecoregions." <https://www.epa.gov/eco-research/ecoregions-north-america>.
- Environmental Protection Agency (EPA).** 2020a. "Compensatory Mitigation Requirements under CWA Section 404." Washington, DC: Environmental Protection Agency. <https://www.epa.gov/cwa-404/background-about-compensatory-mitigation-requirements-under-cwa-section-404>.
- Environmental Protection Agency (EPA).** 2020b. "Ecoregions." Washington, DC: Environmental Protection Agency. <https://www.epa.gov/eco-research/ecoregions>.
- Environmental Protection Agency (EPA).** 2020c. "Overview of Clean Water Act Section 404." Washington, DC: Environmental Protection Agency. <https://www.epa.gov/cwa-404/overview-clean-water-act-section-404>.
- Environmental Protection Agency (EPA), and Army Corps.** 2015. "Clean Water Rule: Definition of 'Waters of the United States'." Washington, DC: Environmental Protection Agency.
- Environmental Protection Agency (EPA), and Army Corps.** 2020. "The Navigable Waters Protection Rule: Definition of 'Waters of the United States'." Washington, DC: Environmental Protection Agency.
- Federal Emergency Management Agency (FEMA).** 2020a. "FIMA NFIP Redacted Claims Data Set." OpenFEMA. <https://www.fema.gov/media-library/assets/documents/180374>.
- Federal Emergency Management Agency (FEMA).** 2020b. "Policy & Claim Statistics for Flood Insurance." <https://www.fema.gov/policy-claim-statistics-flood-insurance>.
- Federal Emergency Management Agency (FEMA).** 2021a. "OpenFEMA Dataset: FIMA NFIP Redacted Claims-v1." https://www.fema.gov/openfema-data-page/_ma-n_p-redacted-claims-v1.
- Federal Emergency Management Agency (FEMA).** 2021b. "OpenFEMA Dataset: FIMA NFIP Redacted Policies-v1." https://www.fema.gov/openfema-data-page/_ma-n_p-redacted-policies-v1.
- FRED Economic Data.** 2021. "Gross Domestic Product: Implicit Price Detor." <https://fred.stlouisfed.org/series/GDPDEF>.
- Gallagher, Justin.** 2014a. "Learning about an Infrequent Event: Evidence from Flood Insurance Take-Up in the United States." *American Economic Journal: Applied Economics* 6 (3): 206–33.
- Gallagher, Justin.** 2014b. "Replication data for: Learning about an Infrequent Event: Evidence from Flood Insurance Take-Up in the United States." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E113898V1>.
- Government Accountability Office (GAO).** 2017. *Flood Insurance: Comprehensive Reform Could Improve Solvency and Enhance Resilience*. Washington, DC: United States Government Accountability Office.
- Ghermandi, Andrea, Jeroen C. J. M. van den Bergh, Luke M. Brander, Henri L. F. de Groot, and Paulo A. L. D. Nunes.** 2010. "Values of Natural and Human-Made Wetlands: A Meta-Analysis." *Water Resources Research* 46 (12). <https://doi.org/10.1029/2010WR009071>.
- Heimlich, Ralph E., Roger Claassen, and Keith Wiebe.** 1998. *Wetlands and Agriculture: Private Interests and Public Benefits*. Washington, DC: US Department of Agriculture, Economic Research Service.
- Homer, Collin, Jon Dewitz, Suming Jin, George Xian, Catherine Costello, Patrick Danielson, Leila Gass et al.** 2020. "Conterminous United States Land Cover Change Patterns 2001–2016 from the 2016 National Land Cover Database." *ISPRS Journal of Photogrammetry and Remote Sensing* 162 (1): 184–99.
- Howard, Peter, and Jeffrey Shrader.** 2019. "An Evaluation of the Revised Definition of 'Waters of the United States'." New York University School of Law, Institute for Policy Integrity. https://policyintegrity.org/documents/Shrader_Howard_Expert_Report_FINAL.pdf.

- Hsieh, Chang-Tai, and Enrico Moretti.** 2019. "Housing Constraints and Spatial Misallocation." *American Economic Journal: Macroeconomics* 11 (2): 1–39.
- Keiser, David A., and Joseph S. Shapiro.** 2019. "Consequences of the Clean Water Act and the Demand for Water Quality." *Quarterly Journal of Economics* 134 (1): 349–96.
- Knutson, Thomas R., John L. McBride, Johnny Chan, Kerry Emanuel, Greg Holland, Chris Landsea, Isaac Held, James P. Kossin, A. K. Srivastava, and Masato Sugi.** 2010. "Tropical Cyclones and Climate Change." *Nature Geoscience* 3 (3): 157–63.
- Kousky, Carolyn, and Erwann Michel-Kerjan.** 2017. "Examining Flood Insurance Claims in the United States: Six Key Findings." *Journal of Risk and Insurance* 84 (3): 819–50.
- Kousky, Carolyn, Sheila M. Olmstead, Margaret A. Walls, and Molly Macauley.** 2013. "Strategically Placing Green Infrastructure: Cost-Effective Land Conservation in the Floodplain." *Environmental Science and Technology* 47 (8): 3563–70.
- Kriesel, Warren, and Craig Landry.** 2004. "Participation in the National Flood Insurance Program: An Empirical Analysis for Coastal Properties." *Journal of Risk and Insurance* 71 (3): 405–20.
- Leibowitz, Scott G., Parker J. Wigington Jr., Kate A. Schofield, Laurie C. Alexander, Melanie K. Vanderhoof, and Heather E. Golden.** 2018. "Connectivity of Streams and Wetlands to Downstream Waters: An Integrated Systems Framework." *Journal of the American Water Resources Association* 54 (2): 298–322.
- Michel-Kerjan, Erwann O.** 2010. "Catastrophe Economics: The National Flood Insurance Program." *Journal of Economic Perspectives* 24 (4): 165–86.
- Mitsch, William J., John W. Day, Li Zhang, and Robert R. Lane.** 2005. "Nitrate-Nitrogen Retention in Wetlands in the Mississippi River Basin." *Ecological Engineering* 24 (4): 267–78.
- Narayan, Siddharth, Michael W. Beck, Paul Wilson, Christopher J. Thomas, Alexandra Guerrero, Christine C. Shepard, Borja G. Reguero, Guillermo Franco, Jane Carter Ingram, and Dania Trespalcacios.** 2017. "The Value of Coastal Wetlands for Flood Damage Reduction in the Northeastern USA." *Scientific Reports* 7 (1): Article 9463.
- Nolte, Christoph.** 2020a. "Data for: High-resolution land value maps reveal underestimation of conservation costs in the United States [Dryad]." <https://doi.org/10.5061/dryad.np5hqbzq9>.
- Nolte, Christoph.** 2020b. "High-Resolution Land Value Maps Reveal Underestimation of Conservation Costs in the United States." *Proceedings of the National Academy of Sciences* 117 (47): 29577–83.
- Omernik, James M.** 1987. "Map Supplement: Ecoregions of the Conterminous United States." *Annals of the Association of American Geographers* 77 (1): 118–25.
- PRISM Climate Group, Oregon State University.** 2021. "PRISM Monthly Precipitation: Recent Years." <https://prism.oregonstate.edu/recent/>.
- Powell, John Wesley, Grove Karl Gilbert, Clarence Edward Dutton, Almon Harris Thompson, and Willis Drummond.** 1879. *Report on the Lands of the Arid Region of the United States: With a More Detailed Account of the Lands of Utah. With Maps.* Washington, DC: US Government Printing Office.
- Science Advisory Board.** 2020. *U.S. EPA, Letter to Andrew Wheeler, 27 February 2020, SAB Commentary on the Proposed Rule Defining the Scope of Waters Federally Regulated under the Clean Water Act, EPA-SAB-20-002.* Washington, DC: United States Environmental Protection Agency.
- Spalding, Mark D., Anna L. McIvor, Michael W. Beck, Evamaria W. Koch, Iris Möller, Denise J. Reed, Pamela Rubinoff et al.** 2014. "Coastal Ecosystems: A Critical Element of Risk Reduction." *Conservation Letters* 7 (3): 293–301.
- Sullivan, S. Mažeika P., Mark C. Rains, and Amanda D. Rodewald.** 2019. "Opinion: The Proposed Change to the Definition of 'Waters of the United States' Flouts Sound Science." *Proceedings of the National Academy of Sciences* 116 (24): 11558–61.
- Sunding, David, and David Zilberman.** 2002. "The Economics of Environmental Regulation by Licensing: An Assessment of Recent Changes to the Wetland Permitting Process." *Natural Resources Journal* 42 (1): 59–90.
- Sun, Fanglin, and Richard T. Carson.** 2020. "Coastal Wetlands Reduce Property Damage During Tropical Cyclones." *Proceedings of the National Academy of Sciences* 117 (11): 5719–25.
- Taylor, Charles A., and Hannah Druckenmiller.** 2022. "Replication Data for: Wetlands, Flooding, and the Clean Water Act." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E153221>.
- Thomas, H., and T. R. Nisbet.** 2007. "An Assessment of the Impact of Floodplain Woodland on Flood Flows." *Water and Environment Journal* 21 (2): 114–26.
- Tiner, Ralph W.** 2003. "Estimated Extent of Geographically Isolated Wetlands in Selected Areas of the United States." *Wetlands* 23 (3): 636–52.
- Turner, R. Kerry, Stavros Georgiou, and Brendan Fisher.** 2008. *Valuing Ecosystem Services: The Case of Multi-functional Wetlands.* New York: Routledge.

- Turner, R. Kerry, Jeroen C. J. M. Van Den Bergh, Tore Söderqvist, Aat Barendregt, Jan Van Der Straaten, Edward Maltby, and Ekko C. Van Ierland.** 2000. "Ecological-Economic Analysis of Wetlands: Scientific Integration for Management and Policy." *Ecological Economics* 35 (1): 7–23.
- US Census Bureau.** 2016b. "TIGER Counties [TIGER/2016/Counties]." https://www.census.gov/geographies/mapping-_les/time-series/geo/tiger-line-_le.html.
- US Geological Survey.** 2021. "National Hydrography Dataset (ver. USGS National Hydrography Dataset Best Resolution (NHD) for Hydrologic Unit (HU) 12. (published 20191002))." <https://www.usgs.gov/core-science-systems/ngp/national-hydrography/access-national-hydrography-products>.
- US Census Bureau.** 2016b. "TIGER Counties [TIGER/2016/Counties]." https://www.census.gov/geographies/mapping-_les/time-series/geo/tiger-line-_le.html.
- US Geological Survey.** 2021. "National Hydrography Dataset (ver. USGS National Hydrography Dataset Best Resolution (NHD) for Hydrologic Unit (HU) 12. (published 20191002))." <https://www.usgs.gov/core-science-systems/ngp/national-hydrography/access-national-hydrography-products>.
- Vanderhoof, Melanie K., and Laurie C. Alexander.** 2016. "The Role of Lake Expansion in Altering the Wetland Landscape of the Prairie Pothole Region, United States." *Wetlands* 36 (2): 309–21.
- Wagner, Katherine.** 2019. "Adaptation and Adverse Selection in Markets for Natural Disaster Insurance." SSRN 3467329.
- Woodruff, Jonathan D., Jennifer L. Irish, and Suzana J. Camargo.** 2013. "Coastal Flooding by Tropical Cyclones and Sea-Level Rise." *Nature* 504 (7478): 44–52.